

From Tracking Infrastructure to AI Profiling

An Unsupervised Clustering Analysis of Online Tracking Practices and Their Ethical Implications

AI Ethics Seminar - Final Assignment | Maxim Lavelin

Abstract

This study examines how online tracking supports AI-driven profiling and what ethical concerns arise from algorithmic inference. Using the March 2025 WhoTracks.me dataset, I constructed a site-level dataset covering 10,000 websites and applied unsupervised machine-learning methods to identify patterns in tracking behavior. K-Means clustering produced two broad regimes: Lower Tracking Intensity and High Tracking Intensity. The results show that online tracking is widespread, highly concentrated, and unevenly distributed across website categories, with News and Portals showing particularly intensive tracking. DBSCAN also identified a group of websites with unusually extreme tracking activity.

The main ethical finding is that the two tracking regimes did not exist in the original dataset; they were created by the clustering algorithm. Although this study classifies websites rather than individuals, the same computational logic can be applied to behavioral data to generate inferred categories, risk scores, and predictive profiles. The study therefore demonstrates that the ethical challenge of AI profiling lies not only in collecting data, but also in transforming behavioral traces into new classifications and knowledge that users did not explicitly disclose and may be unable to anticipate.

1. Introduction

The modern web is built upon an extensive infrastructure of third-party tracking technologies. Website visits activate trackers that collect information about browsing behavior, device characteristics, cookies, and navigation patterns. These data support advertising and analytics while also providing raw material for AI-generated behavioral profiles and predictions (Englehardt & Narayanan, 2016; Zuboff, 2019). Privacy concerns therefore extend beyond collection to the inferences produced from data. AI systems can infer characteristics and future actions that users never explicitly disclosed, making meaningful consent increasingly difficult (Mai, 2016).

This study asks: How do online tracking practices enable AI-driven profiling, and what ethical concerns emerge from algorithmic inference? I apply K-Means and DBSCAN to a site-level dataset engineered from the WhoTracks.me March 2025 release. Empirically, the study measures tracking across 10,000 websites; conceptually, it shows how machine learning creates classifications absent from the original data. Although websites rather than individuals are classified, the computational logic mirrors AI-driven user profiling.

2. Literature Review

2.1 Tracking Infrastructure and Inference

Englehardt and Narayanan (2016) documented an extensive third-party tracking ecosystem based on cookies, browser fingerprinting, and cookie syncing. They also found strong market concentration, enabling a small number of firms to observe behavior across many websites. Mai (2016) argues that the resulting privacy problem is not limited to collection: algorithms transform behavioral traces into new information, including inferred interests, health conditions, political preferences, and financial characteristics. Because such categories may not exist when consent is obtained, users cannot meaningfully anticipate all future inferences.

2.2 Behavioral Prediction and Synthesis

Zuboff (2019) situates this infrastructure within surveillance capitalism. Behavioral data become behavioral surplus that machine-learning systems convert into prediction products about future actions. Together, the three sources describe one process: tracking supplies behavioral signals, algorithmic inference creates new knowledge, and commercial incentives make prediction economically valuable. The

present study operationalizes this framework by testing whether unsupervised learning can generate meaningful tracking regimes from website-level signals.

3. Methodology

3.1 Data and Features

The analysis uses the WhoTracks.me March 2025 dataset, consisting of sites.csv, sites_trackers.csv, trackers.csv, companies.csv, and domains.csv. I constructed a site-level dataset with 10,000 rows. Continuous features captured tracker volume, company diversity, cookie activity, query-string tracking, referer leakage, third-party cookies, and overall tracking prevalence. Facebook presence and one-hot website categories were also included. Google presence was retained descriptively but excluded from clustering because its 99.2% prevalence provided almost no discriminative value. Missing values were median-imputed; continuous variables were standardized, while binary variables remained at 0/1.

3.2 Clustering and Statistical Analysis

K-Means was selected as the main method because its cluster centers provide interpretable profiles. Values of K from 2 to 10 were compared using inertia and silhouette scores. As shown in Figure 1, K = 2 produced the highest silhouette score (0.467), while additional clusters yielded diminishing reductions in inertia. The solution was labeled Lower Tracking Intensity and High Tracking Intensity. DBSCAN was used only for outlier detection, and a chi-square test assessed the association between website category and cluster membership.

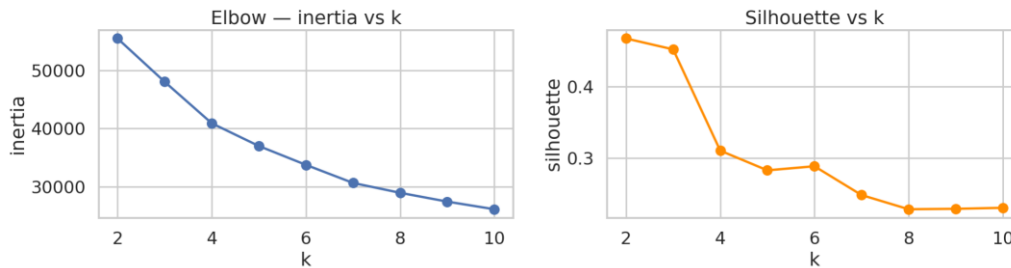


Figure 1. Elbow and silhouette analysis for selecting the number of K-Means clusters.

3.3 Ethical Considerations

Using aggregated website-level data avoided processing individual browsing histories. Feature selection and cluster labels were treated as analytical choices rather than objective truths, and the resulting groups were interpreted descriptively. The same methodology could classify people from behavioral data, illustrating why data minimization, transparent criteria, and human review of inferred categories are necessary before, during, and after model deployment.

4. Results

4.1 Descriptive Results

Across the 10,000 websites, the mean number of trackers was 6.25 and the median was 5.2. Google-affiliated trackers appeared on 99.2% of sites and Facebook/Meta trackers on 48.9%. Tracking varied substantially by category. As shown in Figure 2, News and Portals had the highest mean (10.85), while Government and Banking averaged 3.06 and 3.82 trackers, respectively.

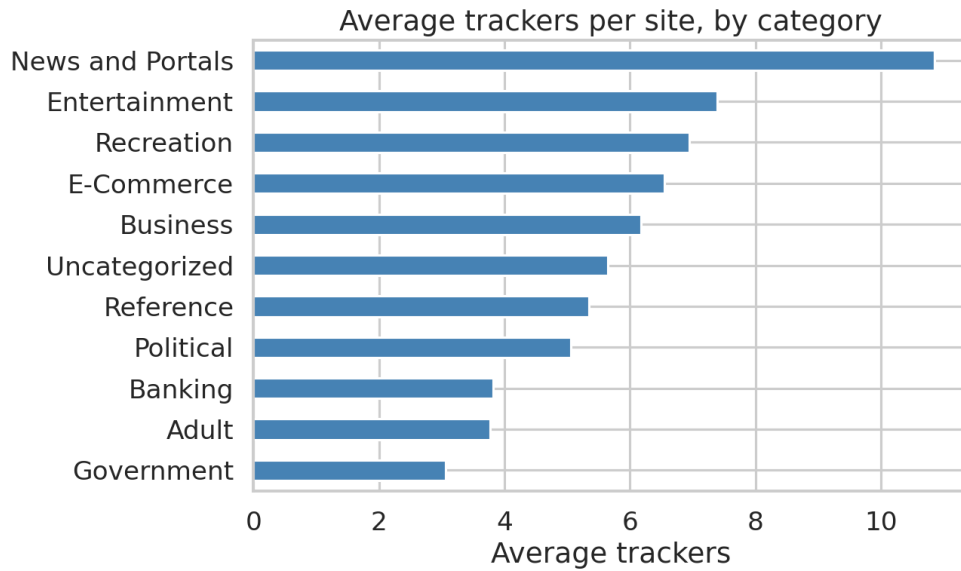


Figure 2. Average number of trackers per site by website category.

4.2 K-Means Cluster Profiles

K-Means identified High and Lower Tracking Intensity regimes. Table 1 shows that the high-intensity group had more than twice as many trackers and companies, substantially higher referer leakage, a higher third-party cookie score, and greater Facebook presence.

Table 1. Mean feature values by tracking regime.

Feature	High Tracking Intensity	Lower Tracking Intensity
Average trackers	12.05	4.87
Tracking companies	9.14	3.61
Referer leakage score	0.608	0.197
Third-party cookie score	0.013	0.003
Facebook presence	76.8%	42.3%

Figure 3 presents the full cluster-center pattern. The silhouette score indicates meaningful but overlapping structure, suggesting a continuum rather than two naturally discrete website types.

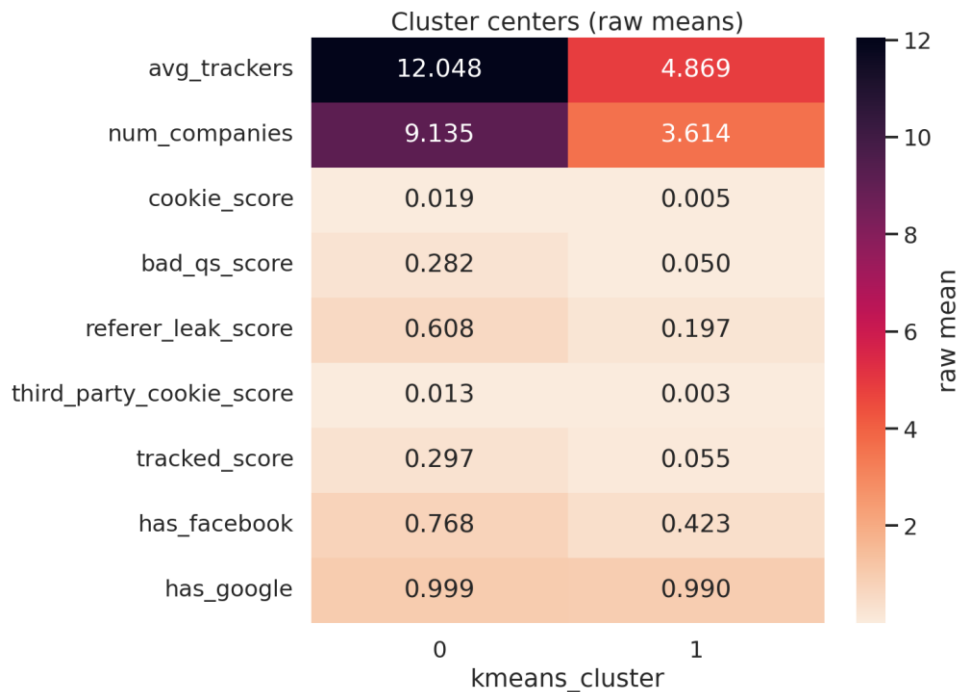


Figure 3. Mean tracking characteristics of the High and Lower Tracking Intensity clusters.

4.3 Category Association and Outliers

The category-cluster association was statistically significant ($p < .001$). Figure 4 shows that News and Portals were strongly overrepresented in the High Tracking Intensity regime (standardized residual = +23.7), while Government, Banking, Adult, and several other categories were underrepresented. DBSCAN identified 634 outliers; the most extreme sites contained approximately 26-35 trackers. Selected cases are reported in Appendix A.

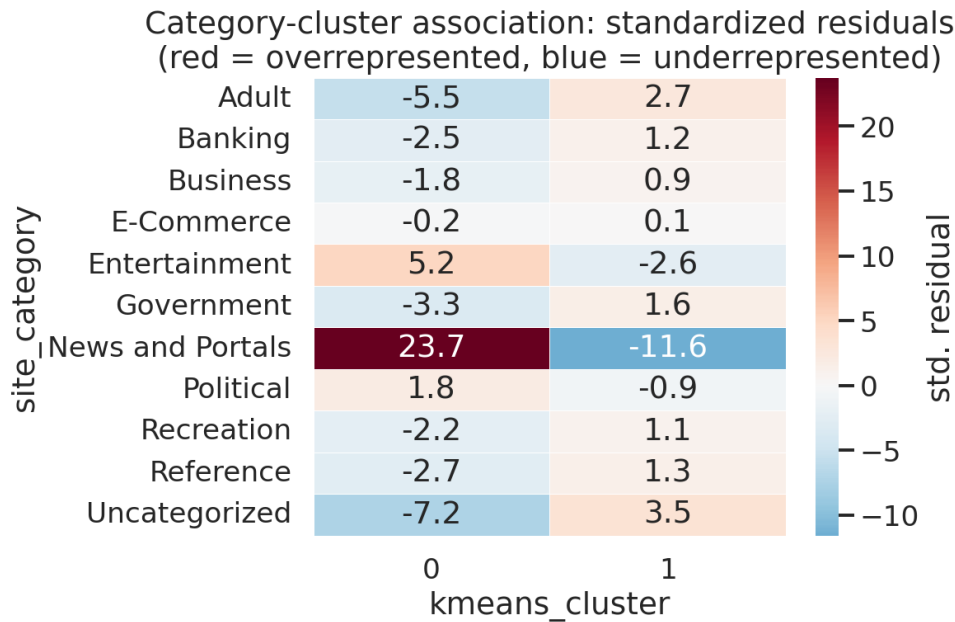


Figure 4. Standardized residuals for the association between website category and tracking regime.

5. Discussion and Conclusion

5.1 Discussion

The 99.2% Google and 48.9% Facebook presence rates are consistent with Englehardt and Narayanan's (2016) description of a tracking ecosystem dominated by a small number of firms. News and Portals averaged 10.85 trackers and were strongly overrepresented in the high-intensity regime. This is consistent with Zuboff's (2019) account of behavioral data being converted into commercial prediction products. A plausible explanation is that advertising-dependent websites have stronger incentives to integrate tracking and audience measurement; however, the observational design establishes association, not causation.

Government and Banking sites occupied the lower end of the distribution, and Adult sites were underrepresented in the high-intensity regime. One possible interpretation is that sectors facing stronger regulatory obligations, reputational risks, or expectations of confidentiality deploy fewer third-party trackers. Because regulation and organizational policy were not directly measured, this interpretation remains tentative.

The cluster labels did not exist in the original dataset; they were generated from statistical relationships among selected features. This illustrates Mai's (2016) argument that privacy harm arises when processing creates new knowledge. Applied to people, similar methods can generate preferences, identities, risk scores, and predictive profiles. The 0.467 silhouette score also shows why these classifications should not be treated as natural or permanent: they depend on feature selection, preprocessing, model choice, and labeling decisions.

5.2 Limitations

The study uses one monthly snapshot and classifies websites rather than users. WhoTracks.me measurements originate from consenting Ghostery and Cliqz extension users (Karaj et al., 2018), who may differ from the wider internet population in privacy awareness, geography, browser choice, and browsing behavior. The direction and magnitude of this sampling bias cannot be determined here. Results also

depend on the selected features and $K = 2$, while strong correlations among several indicators reduce the number of independent signals driving the clusters.

5.3 Conclusion

This study measured web tracking across 10,000 websites and identified two regimes using unsupervised learning. Tracking was extensive and concentrated: Google appeared on 99.2% of sites, Facebook on 48.9%, and News and Portals averaged 10.85 trackers compared with a global mean of 6.25. The clustering algorithm generated classifications absent from the original data. Although the analysis focused on websites, the same logic underlies user profiling. The central ethical challenge therefore lies not only in collecting data, but in transforming behavioral traces into knowledge, classifications, and predictions that users never explicitly disclosed and cannot meaningfully anticipate.

6. References

- Englehardt, S., & Narayanan, A. (2016). Online tracking: A 1-million-site measurement and analysis. In Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (pp. 1388-1401). Association for Computing Machinery. <https://doi.org/10.1145/2976749.2978313>
- Karaj, A., Macbeth, S., Berson, R., & Pujol, J. M. (2018). WhoTracks.Me: Shedding light on the opaque world of online tracking. arXiv. <https://doi.org/10.48550/arXiv.1804.08959>
- Mai, J.-E. (2016). Big data privacy: The datafication of personal information. The Information Society, 32(3), 192-199. <https://doi.org/10.1080/01972243.2016.1153010>
- Zuboff, S. (2019). The age of surveillance capitalism: The fight for a human future at the new frontier of power. PublicAffairs.

7. Appendices

Appendix A. Selected DBSCAN Outliers

DBSCAN identified 634 websites outside the main density structure. Table A1 lists six of the highest-tracking cases. These are tracking outliers, not evidence of malware or malicious activity.

Table A1. Selected highest-tracking DBSCAN outliers.

Website	Average trackers	Tracking companies	Category
gladly.io	35.20	26.71	Uncategorized
adage.com	34.47	25.73	Uncategorized
arkadium.com	33.16	26.98	Uncategorized
athlonsports.com	33.11	26.42	Uncategorized
inquirer.net	32.30	26.56	News and Portals
thestreet.com	31.32	26.38	News and Portals

Appendix B. Additional Diagnostic Figures

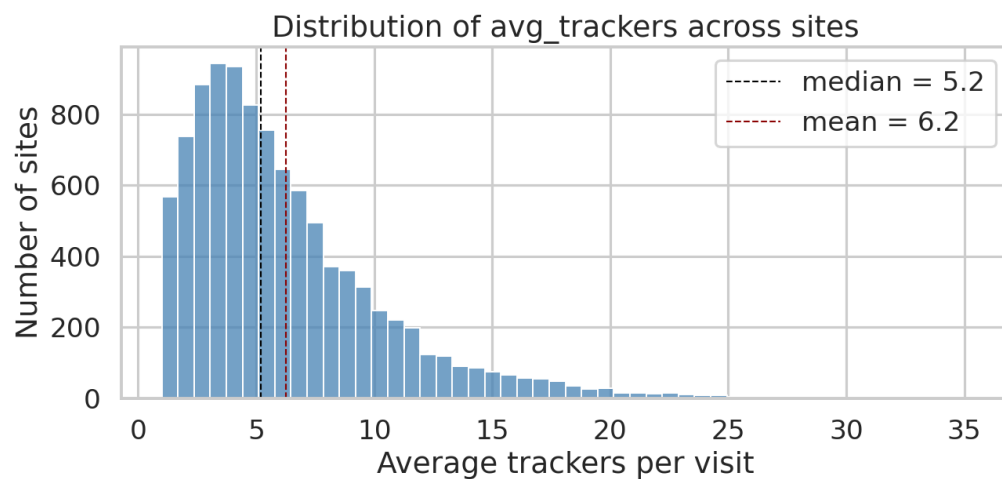


Figure B1. Distribution of average trackers across websites.

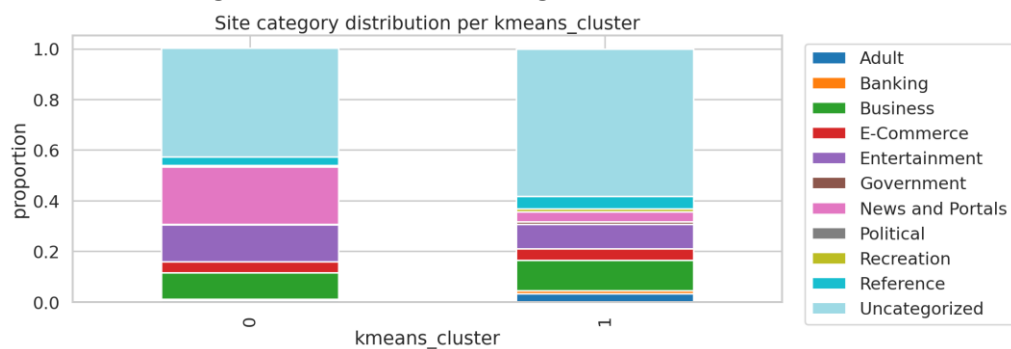


Figure B2. Website-category distribution across K-Means clusters.

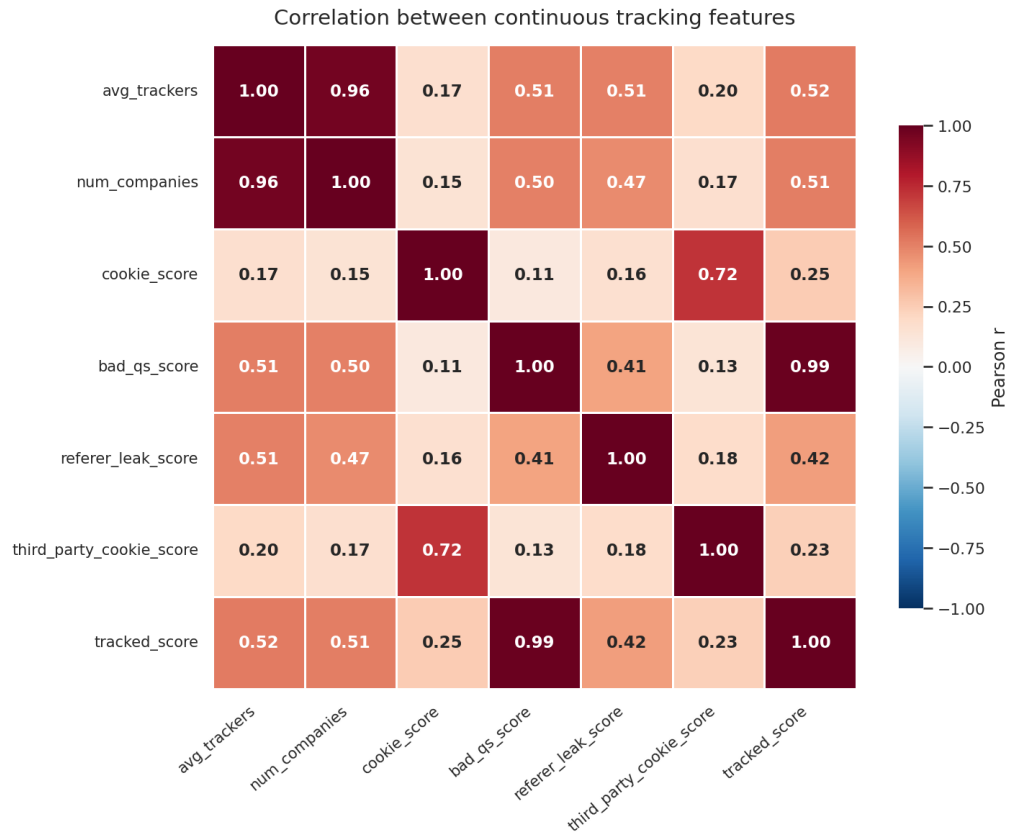


Figure B3. Correlation matrix for continuous tracking features.

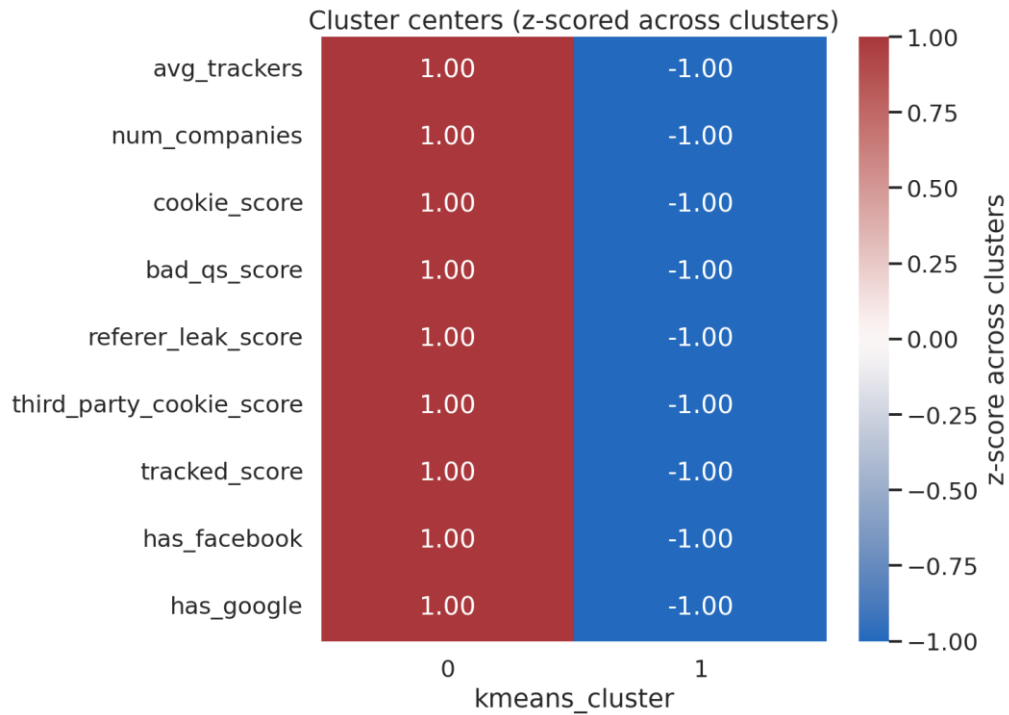


Figure B4. Standardized cluster-center heatmap.

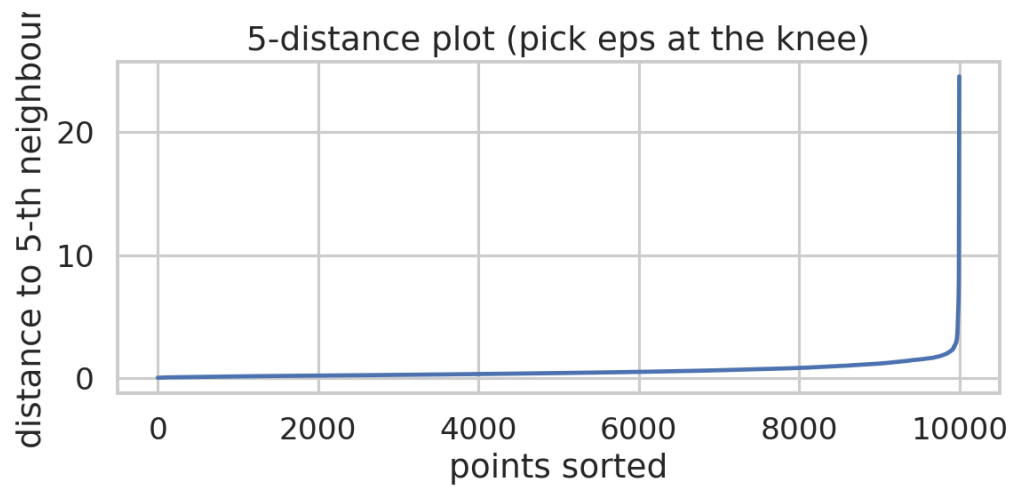


Figure B5. Five-nearest-neighbor distance plot used to guide DBSCAN epsilon selection.
 DBSCAN outliers (1 = outlier, 0 = core) (explained var: 64.85%)

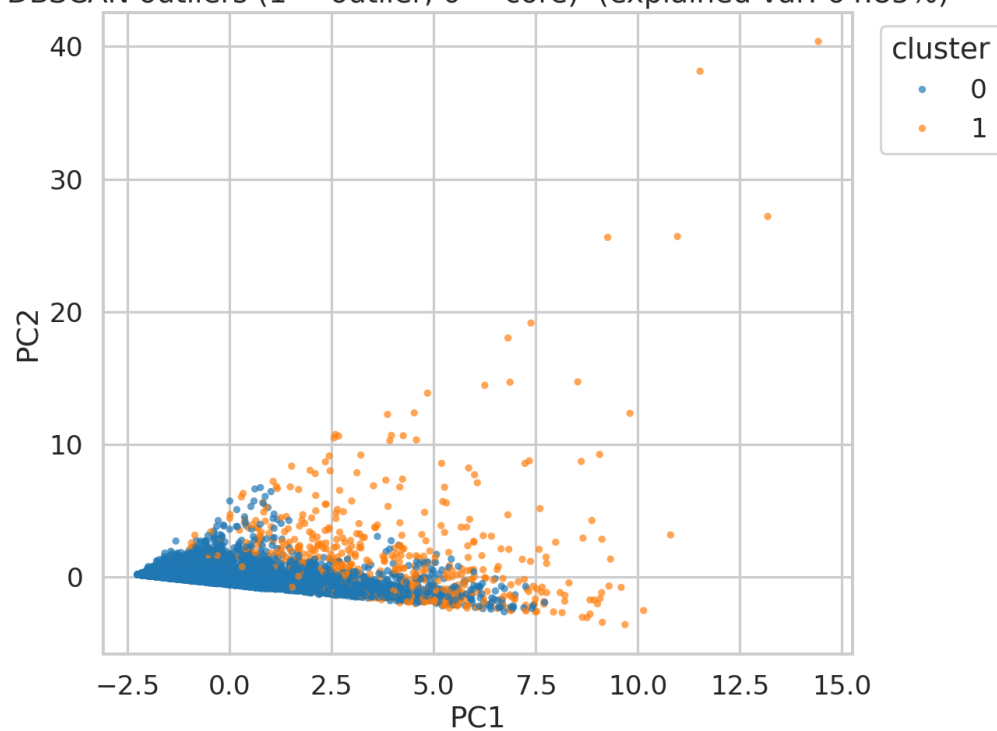


Figure B6. PCA projection of DBSCAN outliers.

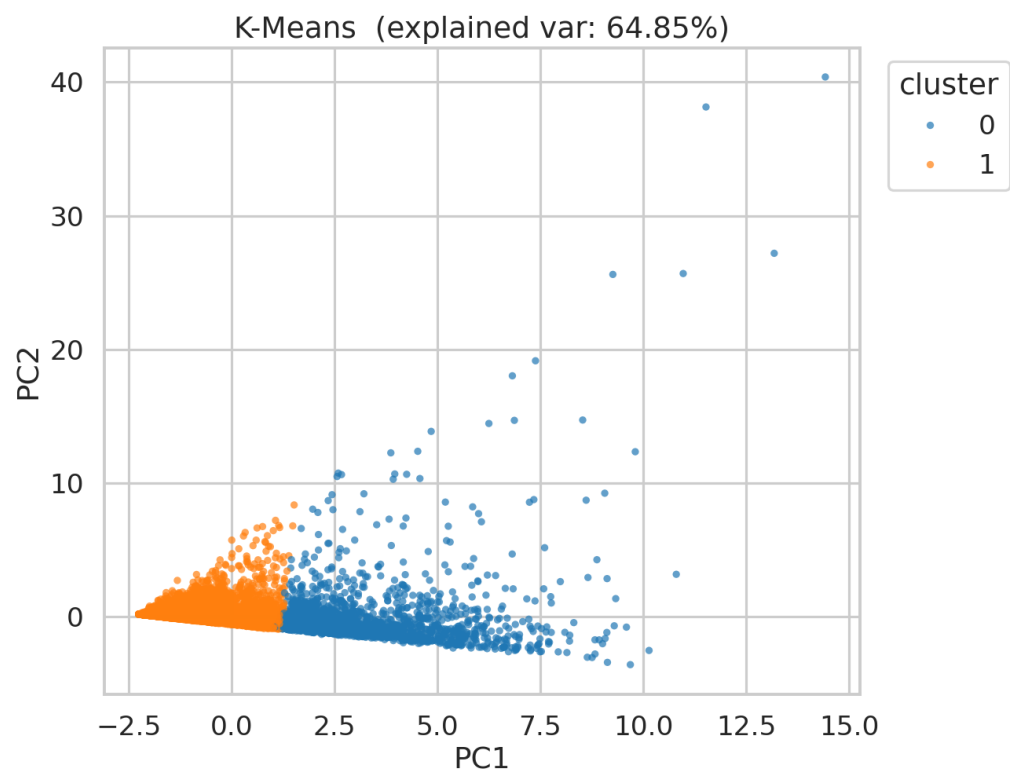


Figure B7. PCA projection of K-Means clusters.